Statistical Signal Processing
Lecture 11:
Adaptive filters and the LMS algorithm

Literature and Problems: See course schedule

Agenda (preliminary):

Intro: the concept of adaptive filtering

Generic model for adaptive filtering
- **FIR filter w(n), input vector x(n), desired signal d(n), output y(n), error e(n)**
- **SLIDE: Application example: adaptive system identification**
- **Cost function to be minimized → w(n)**

SLIDE: Optimal adaptive filter – Wiener Hopf equation
- **"Practical" version: ensemble means → sample means**
- **Complexity**

The principle of Steepest Descent: iterative search towards optimal filter
- **Cost function as function of filter coefficients is (L+1)-dimensional elliptic paraboloide with unique minimum**
- **Filter coefficients updating equation**

The standard LMS algorithm
- **Simplification of cost function**
- **Gradient of simplified cost function**
- **The LMS updating equation for filter coefficients**

SLIDE: The standard LMS algorithm and its performance
- **Stability constraint**
- **Convergence speed**
- **Stationary and excess mean square error**

Variants of the standard LMS algorithm
- **Normalized LMS algorithm: time-varying step size**
- **Leaky LMS algorithm: decreasing algorithm memory**
- **Combination: LNLMS algorithm**

SLIDE: Applications of adaptive filtering
DEMO: Adaptive System Identification
DEMO: Adaptive Linear Prediction and Noise Reduction